

Capstone – Final Project**Name – Balajee Srinivasan****Student No – 2230177****Date – 03/03/2026**

CSEC 594 Security Capstone: Final Project

Phishing Link Scanner Tool

I. Project Overview

Phishing is one of the most widespread cybersecurity risks targeting people and organizations in the present day. Hackers often design fraudulent websites and malicious URLs that masquerade as valid services, including banks, cloud services and social networks websites. These phishing sites are supposed to defraud them into entering information of great sensitivity such as passwords, finances, and any personal information.

This capstone project aims at designing and developing a Phishing URL Detection and Risk Analysis Tool. The system also examines URLs and detects suspicious behavior that is usually used in phishing attacks. Using the indicators identified, the system will determine a risk score and classify the URL as Low Risk, Medium Risk or High Risk.

This tool illustrates the result of using the security applications to analyze the URLs through the use of heuristic detection applications to ascertain possible phishing activities. The project aims at integrating rule-based approach in detection with a user friendly interface, which enables users to test the safety of a URL at an easy location.

II. Initial Draft Implementation.

During the draft stage of the project, it was the main aim to create the development environment and install the initial version of the phishing detection engine working. In the first phase, the following tasks were accomplished as it was described in the draft report:

Virtual Lab Setup

Oracle VirtualBox was used to create a secure testing environment. Ubuntu Linux 24.04 LTS was installed in this virtual machine and will be used as the main development platform. This closed system gave the opportunity to test and develop without interfering with the host operating system.

Python Development Environment.

The python development environment was installed after setting up of the virtual machine. A virtual environment was made in Python to control the project dependencies. A number of Python libraries were installed such as:

- tldextract
- idna
- requests
- streamlit

These libraries facilitate URL parsing, HTTP request and development of future interface.

Initial Phishing Scanner

The initial phishing scanner was programmed in Python. This scanner was executed on the terminal and it examined URLs with a number of heuristic checks.

The tool was able to detect:

- Phishing suspicious words like login, verify, secure, account, update, bank and password.
- Unusually long URLs
- Excessive subdomains on the URLs.
- IP address URLs rather than domain names.
- The domain names which have aroused suspicion include .xyz, .top, .zip, .click, and .gq.

All the indicators contributed to a cumulative risk score which was applied to determine the URL as being Low Risk, Medium Risk, or High Risk.

III. Improvements Made on the basis of Draft Feedback.

Having finished with the drafting step, a number of significant changes were made so that the usability and functionality of the phishing deterring tool could be improved.

Implementation of Web-Based Interface.

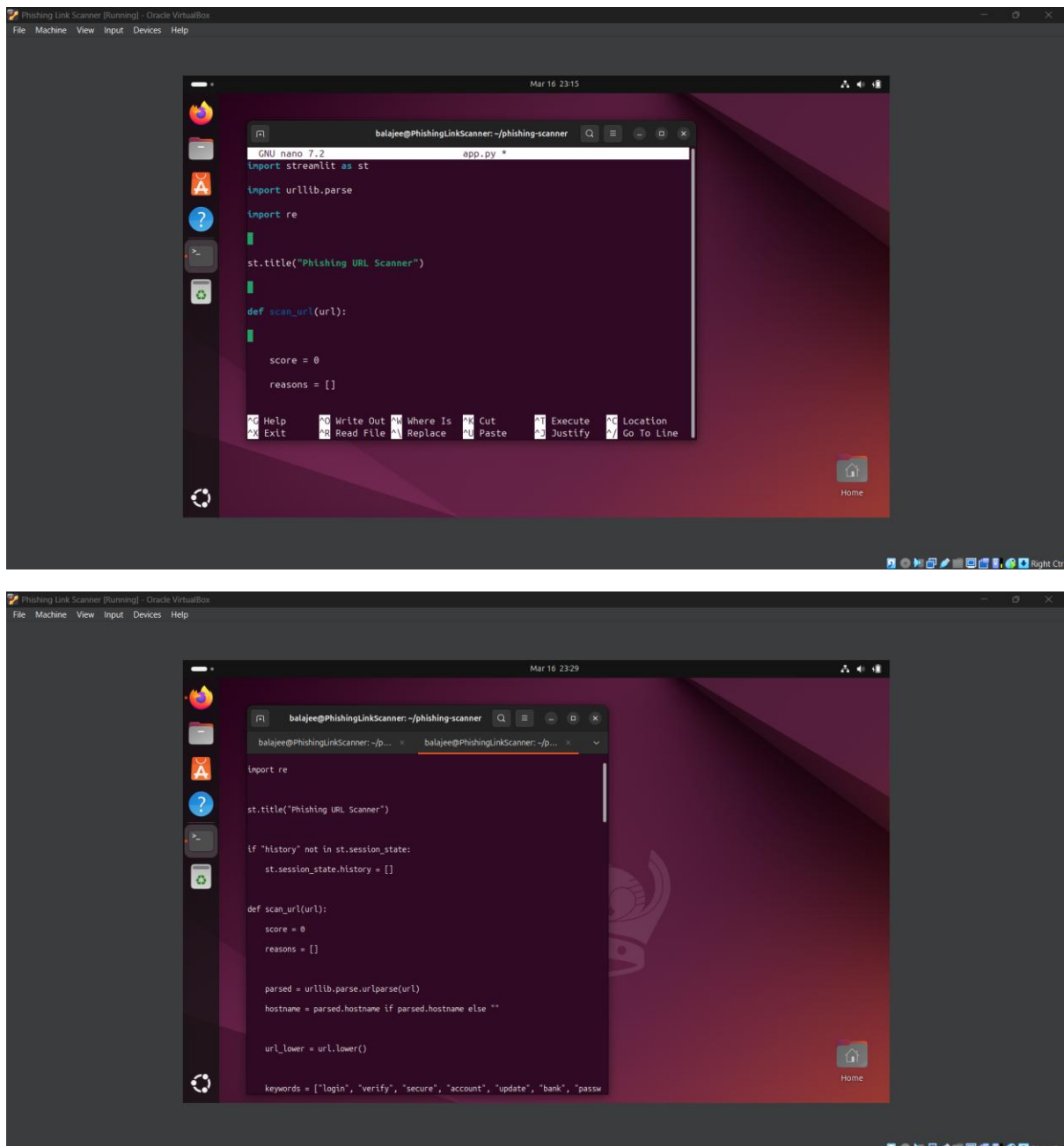
A significant enhancement was turning the command-line scanner into a web-based one with the help of Streamlit. This enabled the phishing scanner to be used via a browser interface and not terminal.

Users can now:

- Enter a URL into a text field
- Click a scan button
- Get a risk analysis result in real-time.

This greatly enhances usability and makes the tool closer to the real life cybersecurity tools.

Screenshot of the code for Website and Scan Results:



Visual Risk Indicators

The other improvement was incorporation of visual risk indicators in the interface.

Rather than the tool presenting bare text outputs, it now presents risk classification by color-coded warnings:

- Green - Low Risk
- Yellow - Medium Risk
- Red - High Risk

Such visual techniques will enable the user to easily grasp the magnitude of the identified risk.

Scan History Feature

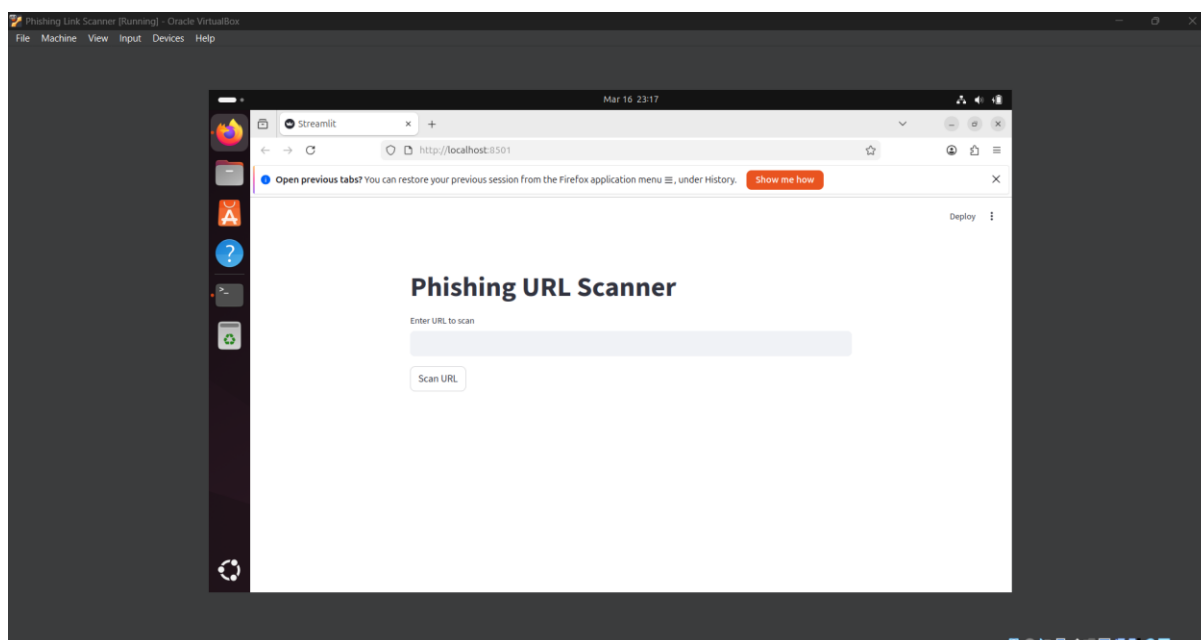
New Scan History dashboard was introduced to monitor the past scanned URLs. The analysis of each URL is saved and presented in a table.

The scan history includes:

URL	Risk Score	Verdict
google.com	0	Low Risk
paypal-login-secure.xyz	20	Medium Risk
secure-bank-update.top	30	Medium Risk

This option enhances the functionality of the tool and enables a user to go over the previously examined URLs.

Screenshot of the final website:



IV. Final System Demonstration

The new system has become web-based and the interface is available at:

<http://localhost:8501>

The person just has to type in a URL and press Scan URL.

Example 1 - Legitimate Website

Input URL:

<https://google.com>

Output:

- Risk Score: 0
- Verdict: Low Risk

The scanner is also accurate in identifying legitimate domains as being safe.

Example 2 – Suspicious URL**Input URL:**

paypal-login-secure.xyz

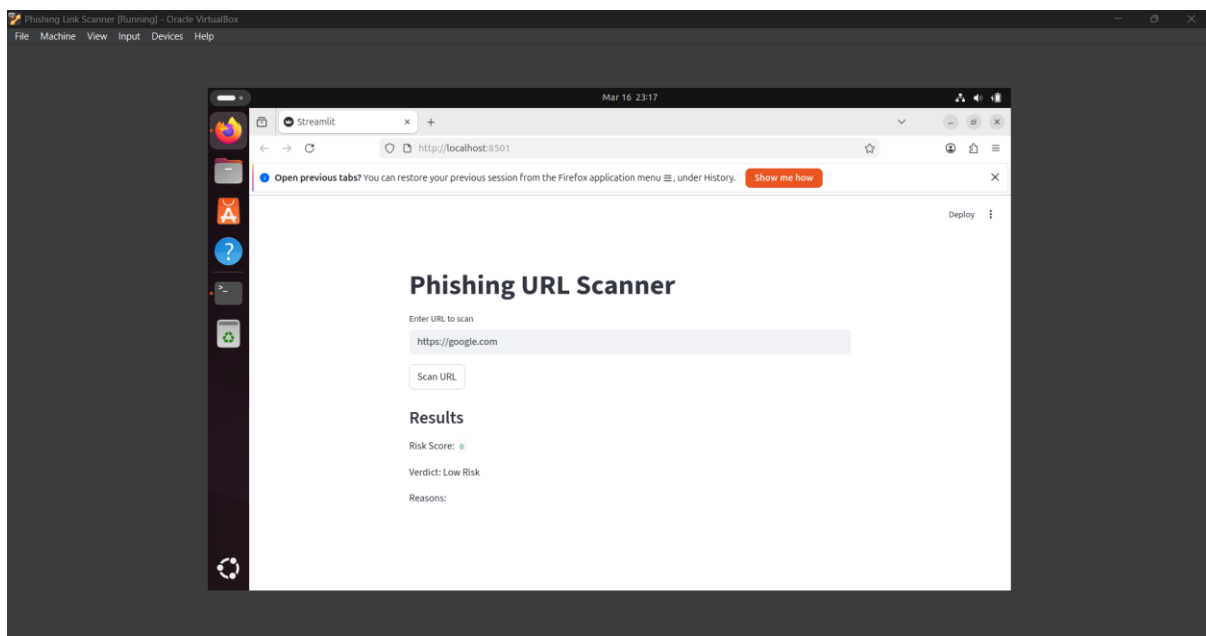
Output:

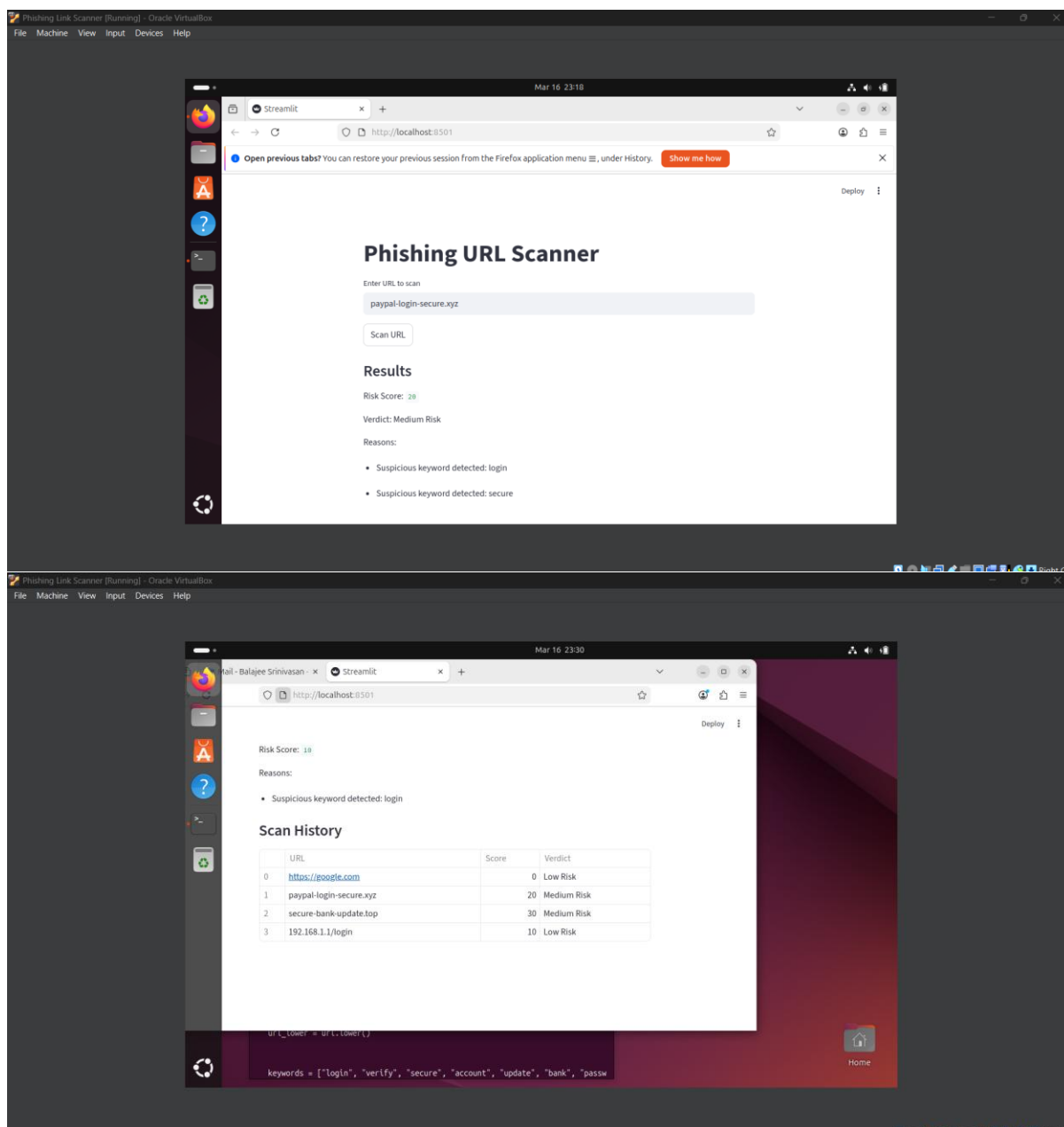
- Risk Score: 20
- Verdict: Medium Risk

Reasons detected:

- Suspicious query found: login.
- Keywords suspected: secure.

These findings are proof of the tool to be effective in detecting phishing features.

Screenshots of Example:



V. Tools and Technologies Used.

The technologies that were applied to create the final version of the project were the following ones:

- **Oracle VirtualBox** - Virtualization platform to develop the testing environment.
- **Ubuntu Linux 24.04 LTS** - Development operating system.
- **Python 3** - The main programming language that is to be utilized in the development of the phishing detection engine.
- **Python Virtual Environment (venv)** - Dependency isolation and package management.
- **Streamlit** - Framework on which the web-based user interface is built.
- **URL Parsing Libraries** - Media of domain structure and URL component analysis.

These technologies enable the tool to be light weighted, portable, and easily extendable.

VI. Final Project Timeline

Phase	Status
Project idea selection	Completed
Virtual lab environment setup	Completed
Python environment configuration	Completed
Initial phishing detection engine	Completed
Web interface implementation	Completed
Scan history feature	Completed
Testing and evaluation	Completed
Final report and documentation	Completed

VII. Final Project achievements were made the following.

The objectives of this project were achieved by the end of it, as follows:

- Building a functional phishing URL detection engine.
- Application of heuristic phishing detection rules.
- Design of a friendly web interface.
- Risk scoring and classification system implementation.
- Implementation of scan history tracking feature.
- Both legitimate and suspicious URLs tested successfully.

The last tool shows the applications of phishing detection systems to analyze the URLs and ensure security information by relying on heuristic analysis methods.

VIII. Conclusion

Through this project, it is possible to exemplify the design and development of a phishing URL detection tool based on Python and Streamlit. The system can detect suspicious URL features, compute risk scores and display the findings in an interactive web interface.

The final system has a number of significant enhancements over the draft version such as a graphical interface, visual risk indicators and a scan history tracker. These additions made the tool very useful and functional.

In general, the project demonstrates the usage of rule based techniques of analysis to identify phishing threats and offers an abundance of future enhancements, including the incorporation of third-party threat intelligence service or machine learning based detecting solutions.